

COURSE OBJECTIVES :

- To understand the python libraries for data science
- To understand the basic Statistical and Probability measures for data science.
- To learn descriptive analytics on the benchmark data sets.
- To apply correlation and regression analytics on standard data sets.
- To present and interpret data using visualization packages in Python.

LIST OF EXPERIMENTS:

1. Download, install and explore the features of NumPy, SciPy, Jupyter, Statsmodels and Pandas packages.
2. Working with Numpy arrays
3. Working with Pandas data frames
4. Reading data from text files, Excel and the web and exploring various commands for doing descriptive analytics on the Iris data set.
5. Use the diabetes data set from UCI and Pima Indians Diabetes data set for performing the following:
 - a. Univariate analysis: Frequency, Mean, Median, Mode, Variance, Standard Deviation, Skewness and Kurtosis.
 - b. Bivariate analysis: Linear and logistic regression modeling
 - c. Multiple Regression analysis
 - d. Also compare the results of the above analysis for the two data sets.
6. Apply and explore various plotting functions on UCI data sets.
 - a. Normal curves
 - b. Density and contour plots
 - c. Correlation and scatter plots
 - d. Histograms
 - e. Three dimensional plotting
7. Visualizing Geographic Data with Basemap

LIST OF EQUIPMENTS: (30 Students per Batch)

Tools: Python, Numpy, Scipy, Matplotlib, Pandas, statmodels, seaborn, plotly, bokeh Note: Example data sets like: UCI, Iris, Pima Indians Diabetes etc.

TOTAL: 45 PERIODS

COURSE OUTCOMES

At the end of this course, the students will be able to:

- Make use of the python libraries for data science
- Make use of the basic Statistical and Probability measures for data science.

- Perform descriptive analytics on the benchmark data sets.
- Perform correlation and regression analytics on standard data sets
- Present and interpret data using visualization packages in Python.

CO's- PO's & PSO's MAPPING

CO / PO	PO1	PO2	PO3	PO4	PO5	PO6	PO7	PO8	PO9	PO10	PO11	PSO1	PSO2	PSO3
CO1	3	3	1	1	3	–	–	–	1	–	2	3	1	3
CO2	3	3	1	2	3	–	–	–	1	–	2	2	-	3
CO3	3	3	2	3	3	–	–	–	1	–	2	2	-	3
CO4	3	2	2	3	3	–	–	–	1	–	3	2	-	3
CO5	3	2	3	3	3	–	–	–	3	–	3	2	-	3
AVG	3.0	2.6	1.8	2.4	3.0	-	-	-	1.4	-	2.4	2.2	1.0	3.0